Thermodynamics For Engineers

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Thermodynamics For Engineers (A05281)

Short Title: Thermodynamics Engs.
Faculty Engineering
Department Engineering Technology
Cluster Engineering
Author CLAIRE.FITZPATRICK
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module takes an analytical approach to the subject of thermodynamics including the areas of system definition, energy and the first law of thermodynamics. The second law of thermodynamics plus entropy will also be covered in this module. Other areas introduced will include: heat engine refrigeration, heat exchangers, energy and the environment. Thermodynamic principles are reinforced by lab experiments.

Programmes

`ICTE-0014`	BEng (Hons) in Mechanical and Manufacturing Engineering (`WD_EMSEN_B`)	2 / 4 / M
`ICTE-0014`	BEng (Hons) in Sustainable Energy Engineering (`WD_CSUEN_B`)	2 / 4 / M
`ENGR-0135`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	2 / 4 / M

Indicative Content

- Thermodynamic laws (zeroth, first and second thermodynamic laws)
- PV cycles and compressed gasses
- Entropy
- Refrigeration and heat engines
- Heat exchangers including plate and concentric tube
- Energy and the environment

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the fundamental elements of gas and thermodynamic laws, properties of pure substances and the effect of energy on our environment.
2. Solve energy calculations using the concepts of thermal efficiency, enthalpy and entropy.
3. Identify refrigeration and heat engine systems and perform analysis of these systems.
4. Demonstrate how heat exchangers of various types operate with their associated calculations.
5. Demonstrate how solar panel is used to calculate solar constant.
6. Demonstrate how heat engine and refrigeration units operate with calculations of efficiency.

Learning and Teaching Methods

- Lectures
- Practicals

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,6
Continuous Assessment	30%	
Lab Report	30%	4,5,6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Cengel, Y.A. _An Introduction to Thermodynamics and Heat Transfer_. 2nd ed. US: McGraw Hill, 2009.

Supplementary Material

- Schmidt, P.S., A.E. Ofodike, J.R. Howell and D.K. Baker. _Thermodynamics: An integrated Learning System_. US: Wiley, 2006.
- Sonntag, R.E. and C. Borgnakke. _Introduction to Engineering Thermodynamics_. 2nd ed. US: Wiley, 2007.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Technology